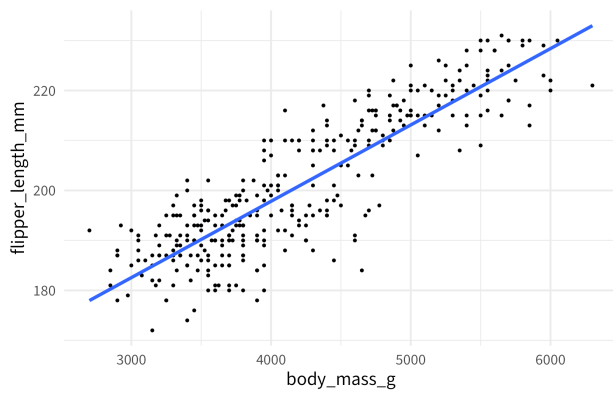
NOTES 07: LINEAR REGRESSION

Stat 120 | Fall 2025

Prof Amanda Luby



$$\widehat{\text{flipper_length_mm}} = 136.73 + 0.02(\text{body_mass_g}) \tag{1}$$

- Intercept (b_0): where does the line cross the y-axis? What is the *prediction* for $x = 0$?
- Slope (b_1): For a 1-unit increase in x, what is the change in the *prediction* for y?

Observation

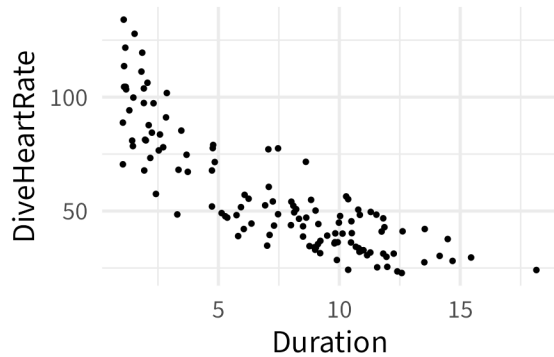
Prediction

Residual

Example: I grab a penguin with a flipper length of 193mm and a body mass of 3475g. What is the **observation**, **prediction**, and **residual** in this case?

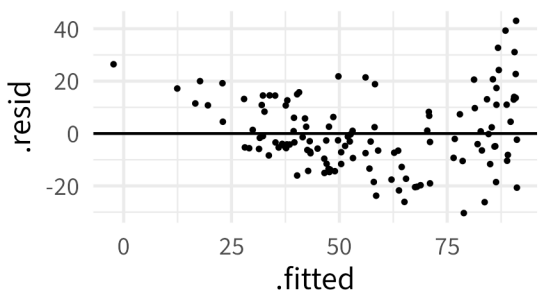
Example: Emperor penguins routinely make dives of 5-12 minutes, with the longest recorded dive over 27 minutes (!!) The rate of oxygen depletion is primarily determined by the penguin's heart rate. `DiveHeartRate` is the bird's heart rate in beats per minute, `Depth` is the depth of the dive (in meters),

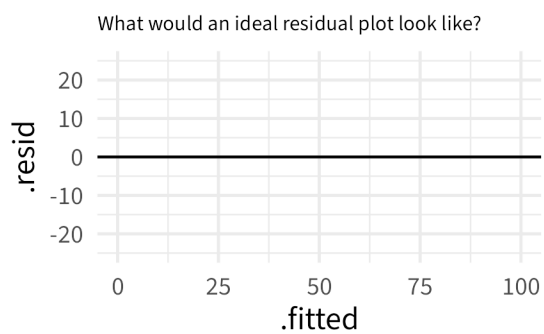
`Duration` is the duration of the dive in minutes, and `Bird` is an ID variable indicating which penguin made the dive.



It can be helpful to look at a *residual scatterplot* when checking for linearity and outliers.

```
dives_lm = lm(DiveHeartRate ~ Duration, data = penguin_dives)
dives_aug = augment(dives_lm)
p1 = ggplot(dives_aug, aes(x = .fitted, y = .resid)) +
  geom_point(size = .5) +
  geom_hline(yintercept =0)
```





Below is the slope and intercept for the transformed model. Write out the linear regression equation and provide an in-context interpretation of the intercept and slope.

```
(Intercept)    Duration
  -0.007926   -0.001817
```